

Lab Exercise 2

Course: COMP08101 Programming for Cyber Security

Week: 2

Topic: Python Programming Basics

Duration: Regular Class Time

Instructor: Raman Singh/ Alan Hart

Submission: N/A

Assessment: N/A

Student Name:

Student ID:

In this lab, you will:

- Use Python variables and basic data types in simple programs.
- Perform arithmetic, comparison and logical operations.
- Use if, elif and else to make decisions.
- Use for and while loops to repeat operations.

Task 01: Write a program that requests the user to enter two values and outputs

- The sum of the 2 values
- The product of the two values
- The second value subtracted from the first value
- The first value divided by the second value
- The first value divided by the second value (integer division)
- The modulus of the first value divided by the second value.
- The first value raised to the power of the second value eg if the first value is 2 and the second value is 3, then your program should output 8 (2^3).

Task 02: Pacific Standard Time (PST) is Greenwich Mean Time (GMT) -8

Central European Time (CET) is GMT + 1

Write a program that asks the user for the time in hours and minutes GMT and outputs the time in the PST and CET zones. Your input should be the time expressed using the 24-hour clock.

Task 03: There are 144 eggs in one gross of eggs and 12 eggs in one dozen eggs. Write a program that accepts a whole number of eggs and outputs the value in terms of gross eggs, dozen eggs and single eggs. NB: This exercise should be done entirely using integer division.

Sample output would be

- Input 160

Output 160 eggs is 1 gross eggs + 1 dozen eggs + 4 eggs

- Input 51 eggs

Output 51 eggs is 0 gross eggs + 4 dozen eggs + 3 eggs

- Input 11 eggs

Output 11 eggs is 0 gross eggs + 0 dozen eggs + 11 eggs

Task 04: Write a Python program that asks the user to enter a positive integer between 1 and 100 inclusive. The program should:

Use if, elif, and else statements to check the user's input.

- If the number is outside the range 1–100, display an appropriate error message.
- If the number is within the range, determine whether it is odd or even using the modulus (%) operator.
- Display an appropriate message for each outcome.

Task 05: Ask the user to enter a number. Use a for loop to display its multiplication table from 1 to 10.

Task 06: Write a program that asks the user to enter a password. Keep asking the user to enter the password while the password is incorrect. When the correct password is entered, display:

Access granted!

Task 07: Write a program which will print out the first 20 terms of the Fibonacci series, where the next term in the sequence is the sum of the previous two terms. (<https://www.mathsisfun.com/numbers/fibonacci-sequence.html>)

0,1,1,2,3,5,8,13,21,34,55,89,144,233,377,610,987,1597,2584,4181

To print on the same line, add a clause to your print statement

```
print(val, end='')
```

Task 08: Security Tool

You are developing a basic security tool for a small organisation. When the program starts, the user must authenticate. After successful authentication, the user is presented with a security menu. The user can then select different security functions. The program should continue running until the user chooses Exit.

The tool will have the following functionalities:

1. Security Event Counter
2. Brute-Force Simulation
3. Login Authentication
4. Exit

The program should perform the following functions:

1. **Security Event Counter:** The program should ask for a password 10 times. There will be a correct password stored in a variable. You need to count the number of times correct and incorrect passwords are entered. Also display the percentage of incorrect passwords entered and raise an alert if this is more than 50%.
2. **Brute-Force Simulation:** Use a predefined password. Allow a maximum number of attempts. Ask the user for a password guess. Count attempts. Stop when the correct password is entered. Stop when the maximum number of attempts is reached.
3. **Login Authentication:** Ask for a username and password. Check the credentials. Display whether authentication succeeded or failed. If it fails, ask for the username and password again. Allow three failed logins. If failed logins reach the maximum attempts, display a message that the account is frozen for 24 hours.
4. **Exit:** If a user selects exit, display a message that says, " Thank you for using the security tool".